

We can approximate this guided distribution accordingly with an unconditional and a conditional prediction model, i.e.,  $p_\theta(\mathbf{a}_0|c, \mathbf{a}_t, t)^\gamma \cdot p_\theta(\mathbf{a}_0|\mathbf{a}_t, t)^{1-\gamma} \approx \tilde{q}(\mathbf{a}_0|c, \mathbf{a}_t)^\gamma \cdot \tilde{q}(\mathbf{a}_0|\mathbf{a}_t)^{1-\gamma}$ . Taking the logarithm, we obtain

$$\log(p_\theta(\mathbf{a}_0|c, \mathbf{a}_t, t)^\gamma \cdot p_\theta(\mathbf{a}_0|\mathbf{a}_t, t)^{1-\gamma}) = \gamma \log p_\theta(\mathbf{a}_0|c, \mathbf{a}_t, t) + (1 - \gamma) \log p_\theta(\mathbf{a}_0|\mathbf{a}_t, t).$$

## C Dataset generation

Here we provide details about the training dataset Alex-MP-20 and the reference dataset Alex-MP-ICSD used throughout this work.

### C.1 Data sources

We obtained crystal structures via three sources:

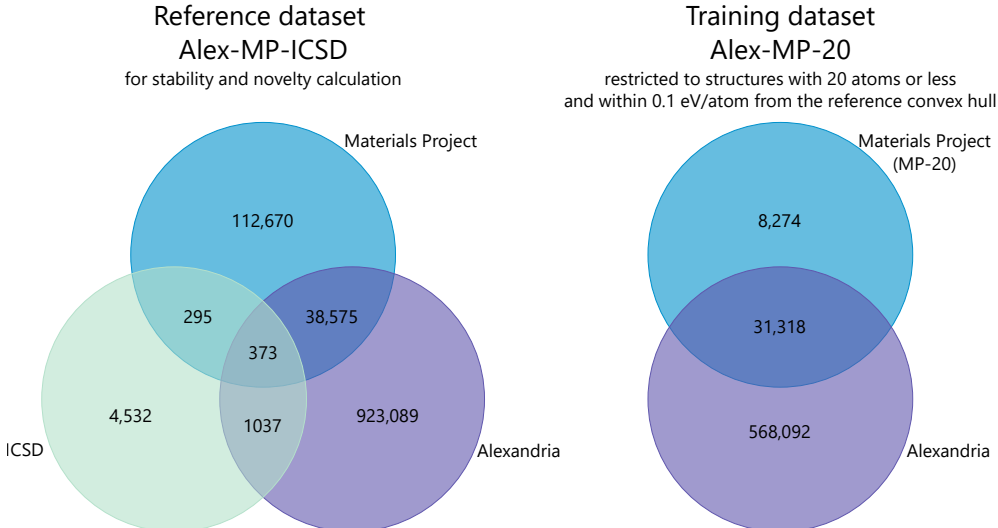
- MP (v2022.10.28, Creative Commons Attribution 4.0 International License) [15], an open-access resource containing DFT-relaxed crystal structures obtained from a variety of sources, but largely based upon experimentally-known crystals.
- The Alexandria dataset [72–74] (Creative Commons Attribution 4.0 International License), an open-access resource containing DFT-relaxed crystal structures from a variety of sources, including a large quantity of hypothetical crystal structures generated by ML methods or other algorithmic means.
- ICSD (release 2023.1) [75], a proprietary database containing crystal structures refined from experiment. For the purposes of dataset generation, we queried ICSD only for experimental crystal structures that were not tagged as already included in MP, and that were directly calculable by DFT (i.e., ordered crystals).

For crystal structures from MP, we retrieved existing calculations via the MP API. For other data sources, we performed new calculations using MP settings to guarantee consistency of data (see Appendix C.2). We then followed MP’s data analysis approach as implemented in `emmet` [76], which includes the following steps:

1. Validation of each individual DFT calculation to ensure required minimum quality criteria are met.
2. Grouping of calculations of equivalent crystal structures, which de-duplicates crystal structures when the same crystal is present in multiple data sources. See Fig. C2 for an overview of the resulting statistics.
3. Application of an empirical correction scheme [77] to address known systematic errors from the Perdew–Burke–Ernzerhof (PBE) functional.
4. Construction of convex hull phase diagrams for each chemical system.

This process resulted in a reference dataset of 1,081,850 unique structures with associated energy above hull values calculated using DFT. We refer to this as the Alex-MP-ICSD dataset. This dataset was then used to derive the Alex-MP-20 dataset, whose element distribution is shown in Fig. C3. The Alex-MP-ICSD dataset is used as a reference for the computation of stability (i.e., energy above hull) and uniqueness of generated structures. To train MatterGen, we employ a subset of the Alex-MP-ICSD dataset, selecting only structures with up to 20 atoms and whose energy above hull

is below 0.1 eV/atom; we refer to this as the Alex-MP-20 dataset. Furthermore, we manually exclude from the Alex-MP-20 dataset all structures belonging to the “well-explored” chemical systems, as defined in Appendix D.4. Additionally, we reserve structures present only in ICSD for testing purposes, and therefore exclude them from the Alex-MP-20 dataset. We report in Fig. C2 the structure provenance and quantity for the reference (Alex-MP-ICSD) and training (Alex-MP-20) datasets. Finally, the dataset employed to train the MatterGen-MP model contains structures from the MP-20 dataset (containing structures with up to 20 atoms) whose energy above hull is below 0.1 eV/atom from the reference convex hull. This is also highlighted in Fig. C2 (right panel, blue circle).



**Fig. C2:** Venn diagrams (not to scale) showing the overlap of crystal structures from the three data sources used in this work, for the reference Alex-MP-ICSD dataset (left) and the Alex-MP-20 training dataset (right). Crystal structures were de-duplicated after calculation and therefore the overlap in this diagram shows cases where the same crystal structure was present in multiple data sources. Note that the statistics for ICSD include only the structures sourced from the ICSD in this study, and not the full ICSD database.

## C.2 DFT details

All DFT calculations were performed using the Vienna ab initio simulation package (VASP) [78] within the projector augmented-wave formalism via `atomate2`, `pymatgen` and `custodian` [79]. All parameters of the calculations were chosen to be consistent with MP [15], including use of the PBE functional [80] and Hubbard U corrections. Specifically, the following workflows were used for the calculation of associated properties:



## D Results

This section contains supplementary information for the results detailed in Section 2.

### D.1 Common experimental details

In this section we provide information about settings used across different experiments, including training and sampling details of MatterGen, and additional information about the MLFF we employ. Details specific to certain experiments are deferred to Appendices D.3 to D.7.

#### D.1.1 Hyperparameters for training base model

The base unconditional model was trained for 1.74 million steps with a batch size of 64 per GPU over eight A100 GPUs using the Adam optimizer [82]. The learning rate was initialized at 0.0001 and was decayed using the ReduceLROnPlateau scheduler with decay factor 0.6, patience 100 and minimum learning rate  $10^{-6}$ . For the training loss we use  $\lambda_{\text{coord}} = 0.1$  and  $\lambda_{\text{cell}} = \lambda_{\text{types}} = 1$ , as well as the recommended value  $\lambda_{\text{CE}} = 0.01$  for the D3PM cross-entropy loss.

#### D.1.2 Hyperparameters for fine-tuning models

For fine-tuning the base model towards different properties, we used a global batch size of 128 and the Adam optimizer. Gradient clipping was applied by value at 0.5. The learning rate was initialized at  $6 \times 10^{-5}$  and we used the same learning rate scheduler as that for the base model. The training was stopped when the validation loss stopped improving for 100 epochs, which resulted in 32,000 - 1.1 million steps depending on the dataset.

#### D.1.3 MatterGen sampling parameters

For both unconditional and conditional generation, we discretized the reverse diffusion process over the continuous time interval  $[0, 1]$  into  $T = 1000$  steps. For each time step, we use ancestral sampling (according to Eqs. (A6), (A8) and (A16)) to sample  $(\mathbf{X}_{t-1}, \mathbf{L}_{t-1}, \mathbf{A}_{t-1})$  given  $(\mathbf{X}_t, \mathbf{L}_t, \mathbf{A}_t)$  using the score model described in Appendix A.8. After each predictor step, one corrector step was applied. We used the Langevin corrector (see Algorithms 4 and 5 in [43]) for the coordinates  $\mathbf{X}_t$  and the lattice  $\mathbf{L}_t$  with signal-to-noise ratio parameters 0.4 and 0.2, respectively.

#### D.1.4 Machine learning force field (MatterSim) details

We used an MLFF trained on 1.08 million crystalline structures which employed the M3GNet [23] architecture with three graph convolution layers and had in total 890,000 parameters. To compute the energy above hull, we used the energy correction scheme compatible with MP (i.e., `MaterialsProject2020Compatibility` from `pymatgen` [79]). Further details on the MLFF will be provided in a separate publication [53].

## D.2 Qualitative analysis of generated structures

Assessing the quality of generated crystals is difficult. In this work, we have used computational metrics to assess the quality of generated materials. However, additional human review of generated materials can be useful to identify failure modes not captured by these metrics. We perform this human analysis of generated crystals in Appendices D.3.2, D.6.2 and D.7.2.

Ultimately, it must at least be possible to create the material in a laboratory setting for any hypothetical material to be of practical use, i.e., the material must be synthesizable. Although it is not practically possible to conclusively determine that a material is synthesizable using theoretical and computational evidence alone, we can use computation as a guide. Throughout this work we use the energy above the convex hull, calculated at 0 K under 0 GPa applied stress, as a signal [83] that a material might be stable at ambient temperature and pressure. We acknowledge that there are additional calculations we could perform (such as computing the phonon spectra) that would improve upon the approximations we make in this work, but robustly assessing synthesizability of a hypothetical crystal structure is still an open research question.

Energy above hull alone is not sufficient, since many metastable materials with finite energy above hulls (“off-hull”) are routinely synthesized, while many on-hull materials have not been, despite best efforts. While an attempt [84] has been made to suggest reasonable threshold values for energy above hull, this is very dependent on chemical system: while some materials (such as carbides, nitrides) are able to tolerate very high energies above convex hull (0.1–1 eV), other materials (such as intermetallics) are only able to tolerate a very small degree of metastability (meV). It is also known [85] that traditional methods of simulation using DFT will give inaccurate energies for some materials, even after empirical corrections [77] are applied which can correct some of the better understood systematic errors. Furthermore, the energy above hull is a measure that is only meaningful if the particular chemical system has been well-explored: for unexplored or partially-explored chemical systems, the energy above hull might be inaccurate simply due to more energetically-favourable phases being unknown.

Using empirical priors to assess the synthesizability of structures is traditionally done by hand by domain experts and is therefore highly dependent on the chemical intuition and background knowledge of a particular system by the expert. It becomes more difficult for a scientist to evaluate a crystal structure picked “at random”, especially as the number of elements increases, as they are less likely to have encountered this material in their prior work. Some materials are hard to analyze in this way because the prior is simply not known or may be biased; for example, any distribution of a specific property calculated from crystalline materials that have already been synthesized will include bias by not simply including the properties of materials that have not yet been synthesized. These biases can be because certain elements are more abundant, cheaper, or easier to process, or because certain materials have garnered more technological interest, rather than because of an *a priori* physical reason why those materials could not be made. Simply put, we do not yet know what the distribution of “possible” crystal structures looks like, even within certain constraints (e.g., maximum primitive cell size or number of elements). Therefore, assessing synthesizability